TaskFlow

Plan • Organize • Achieve

Non-Functional Requirements

Personal & Collaborative Project Management Platform

PROGRAM	**DEPI**
DOCUMENT	**Non-Functional Requirements**
VERSION	**1.0**
PREPARED	**15 October 2026**
STATUS	**Draft for team review and lecturer feedback**
AI	**Out of scope**

> **Target note:** Numeric values in this document are proposed acceptance targets for a DEPI student project. They are not measured results, official DEPI thresholds, or production guarantees. Confirm targets with the team and lecturer, then record actual test conditions and results in the final test report.

1. Purpose

This document defines how well TaskFlow should operate, rather than listing the individual features it provides. It sets proposed criteria for performance, security, usability, reliability, compatibility, maintainability, accessibility, and operational readiness.

TaskFlow serves individual users, freelancers, teams, and companies. Its data may include personal tasks, workspace/project records, time entries, comments, attachments, deliverable reviews, and client-visible progress. Non-functional requirements therefore need to protect both ordinary project data and information shared selectively with clients.

2. Scope and Assumptions

- Requirements apply to the authenticated TaskFlow application and its APIs, database operations, real-time updates, file-handling paths, and optional client portal.
- Public marketing pages are handled separately by Mostafa and are not included in the internal application-module allocation, although shared accessibility and responsive-design principles may also be applied to them.
- The project will not include AI functionality. Health alerts, automation, search, and reports must use explicit rules and stored data.
- Performance depends on the testing environment, dataset, hosting resources, and network conditions. Every reported measurement must state those conditions.
- Azure hosting is planned if available and approved; otherwise, local or an approved alternative environment may be used for demonstration. Any environment-specific limitation must be documented.

3. Priority Definitions

- **Must (P0):** Required for a safe, usable core release.
- **Should (P1):** Important quality target; implement or validate when feasible within the project schedule.
- **Could (P2):** Desirable improvement that may be deferred with the team's agreement.

These priorities are proposed and should be confirmed by the team and lecturer.

4. Performance Requirements

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-PERF-01	Must	Common application screens shall load within an acceptable time under the agreed test conditions.	95th percentile ≤ 2 seconds for common read operations (dashboard, project overview, Kanban board, task details) with a documented representative dataset and up to 20 concurrent test users.	Repeatable load/performance test; record dataset size, request count, environment, and percentile.
NFR-PERF-02	Must	Common data-changing operations shall provide timely feedback.	95th percentile ≤ 3 seconds for common create/update/status-change operations under the same test conditions, excluding file upload time and external provider delay.	Measure request duration for task creation, task update, status change, comment creation, and timer actions.
NFR-PERF-03	Should	Authorized connected users should receive relevant real-time updates quickly.	Update visible to a connected authorized client within 2 seconds in the documented local/demo environment under normal conditions.	Timestamp the successful server-side change and its display on a second connected client.
NFR-PERF-04	Should	Search and filtered lists should remain useful with a representative sample of records.	A typical search/filter request returns within 2 seconds at the agreed test dataset size.	Test common query combinations over a documented set of projects and tasks.
NFR-PERF-05	Should	Reports should complete within a reasonable time without unnecessarily loading unrelated data.	Common project/time reports return within 3 seconds for the agreed sample dataset.	Record execution time and verify the report result is correct.
NFR-PERF-06	Must	The application shall avoid unbounded result lists for large collections.	Lists use pagination, incremental loading, or an explicitly limited result count for records that may grow substantially.	Inspect and test task, project, activity, notification, and time-entry lists.
NFR-PERF-07	Should	Database access should avoid unnecessary repeated queries and large unneeded object graphs.	Performance review identifies and addresses obvious N+1 query patterns and unnecessary data loading in critical screens.	Inspect EF Core queries/logging and profile selected workflows.

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-PERF-08	Could	The system may cache safe, frequently requested reference data where useful.	Any caching introduced must have a documented invalidation rule and must not expose data across users or projects.	Test cache invalidation and cross-user/project isolation.

5. Security Requirements

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-SEC-01	Must	Authentication and account management shall use ASP.NET Core Identity or an equivalent approved secure implementation.	No plaintext passwords or custom reversible password storage; standard secure password hashing and sign-in protections are used.	Review Identity configuration and test sign-in, sign-out, and password-recovery flows.
NFR-SEC-02	Must	Server-side authorization shall protect every workspace, project, task, time entry, attachment, report, client record, and deliverable operation.	100% of defined role/access test cases pass; zero successful unauthorized cross-workspace or cross-client accesses in the release test suite.	Positive and negative tests using separate accounts and direct URL/API requests.
NFR-SEC-03	Must	The system shall enforce least privilege for Owner, Admin, Member, Viewer, and Client access.	Each defined role can perform its permitted actions and is denied actions outside its permissions.	Role-permission matrix and automated/manual authorization tests.
NFR-SEC-04	Must	The application shall validate and safely handle user input.	No known unresolved critical/high-severity input-handling vulnerability remains at release.	Validate server-side model rules and test representative SQL injection, cross-site scripting (XSS), and invalid-input cases.
NFR-SEC-05	Must	State-changing web requests shall be protected against cross-site request forgery (CSRF) where applicable.	Invalid/missing anti-forgery tokens are rejected for applicable authenticated state-changing form requests.	Security tests for MVC forms and state-changing endpoints.

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-SEC-06	Must	Deployed traffic shall use HTTPS.	All application pages and protected endpoints on the deployed environment use HTTPS; HTTP requests are redirected or rejected according to deployment configuration.	Inspect deployment configuration and verify browser/network behavior.
NFR-SEC-07	Must	Credentials, connection strings with passwords, API keys, and other secrets shall not be committed to GitHub.	No active secret is present in tracked source/configuration; deployment secrets are provided through environment configuration or a suitable secret store.	Repository review and secret scan before release.
NFR-SEC-08	Must	File uploads shall be validated and downloads shall require authorization.	Disallowed file types/sizes are rejected, and users cannot retrieve private files by guessing or reusing a URL.	Test allowed/disallowed files, size limits, content validation, and direct-download access.
NFR-SEC-09	Must	Error messages shall not reveal stack traces, secrets, passwords, tokens, or sensitive internal configuration to ordinary users.	Production/demo-facing error responses contain a safe message and a trace/correlation reference where available; detailed diagnostics are kept in restricted logs.	Trigger representative server errors and review responses/log access.
NFR-SEC-10	Must	Sensitive operations and important permission/review changes shall be auditable where defined by the requirements.	Audit records include the actor, action, affected record, and timestamp for the agreed auditable event set.	Execute sample operations and verify corresponding audit entries.
NFR-SEC-11	Should	Sessions shall be managed securely.	Sign-out invalidates the active application session; session/cookie settings follow secure defaults appropriate to the deployment.	Test sign-out, cookie flags, and session behavior on the deployed environment.
NFR-SEC-12	Should	The application shall follow a basic secure-development checklist before release.	No known unresolved critical/high-severity security defect remains in the release checklist.	Review security issues and document checks performed and limitations.

6. Usability Requirements

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-USE-01	Must	New users shall be able to begin using TaskFlow without first creating a team.	An individual user can register, reach a personal workspace, create a project, and create a task without inviting anyone.	Run the first-use journey with a clean account.
NFR-USE-02	Must	Core navigation and terminology shall be consistent across modules.	Shared concepts such as Project, Task, Status, Priority, Timer, and Deliverable use consistent labels and actions.	UI review and task-based walkthrough.
NFR-USE-03	Must	Forms shall identify required fields and explain validation errors clearly.	Invalid form submissions identify the field/problem and preserve valid entered data where practical.	Test missing, invalid, and boundary-value inputs.
NFR-USE-04	Must	The core interface shall support responsive layouts.	Primary journeys remain usable at agreed viewport widths, including a small mobile viewport of approximately 360 CSS pixels and a typical desktop viewport, without unintended horizontal overflow.	Browser responsive-mode tests and device-size screenshots.
NFR-USE-05	Should	Users should be able to complete core workflows with limited assistance.	At least 80% of participants in a small usability test complete the defined core journey without step-by-step assistance; report actual sample size and limitations.	Observe representative users completing project/task creation and, if in scope, client deliverable review.
NFR-USE-06	Should	The interface shall provide useful empty, loading, success, warning, and error states.	Every major list/dashboard has a clear empty state; important user actions show a success or failure result.	UI checklist and walkthrough of first-use and failure scenarios.
NFR-USE-07	Must	Status and priority shall not be communicated by color alone.	Status badges and alerts include text or another accessible label in addition to color.	UI review for task, notification, and project-health states.

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-USE-08	Should	Common actions should minimize unnecessary navigation.	Users can open task details from My Work, Kanban, search, and relevant notifications without losing necessary context.	Task-based usability review.
NFR-USE-09	Should	The visual identity shall be consistent across the authenticated application.	Components use the agreed TaskFlow palette, typography, spacing, and button conventions; the agreed primary palette is documented in shared styles.	UI review across all module pages.
NFR-USE-10	Should	Destructive or irreversible actions shall be clearly communicated.	Deleting, removing access, or other destructive actions require confirmation or an equivalent safe flow; archive is used where appropriate.	Test confirmation and cancellation flows.

7. Reliability and Data Integrity Requirements

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-REL-01	Must	Persisted user changes shall remain available after navigation, refresh, and normal re-login.	All successful persistence test cases retain their saved data; failed writes are not displayed as successful.	Create/update records, refresh/re-login, then compare stored values.
NFR-REL-02	Must	Related database changes shall remain consistent.	Operations that require multiple related writes use a transaction or equivalent consistency mechanism; failure leaves no partial invalid state.	Inject/trigger representative failures and inspect resulting records.
NFR-REL-03	Must	Timer sessions shall produce correct, non-duplicated durations under the defined rules.	All deterministic timer tests match the expected duration within 1 second; no paused interval is counted and no session is counted twice.	Test start/pause/resume/stop, refresh, reconnect, duplicate requests, and incomplete sessions.

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-REL-04	Must	Task updates and real-time messages shall preserve the saved server state as the source of truth.	A failed real-time update does not lose the underlying saved change; after reconnect or refresh, the client can retrieve the current saved state.	Simulate disconnection and verify data after reconnect/refresh.
NFR-REL-05	Must	Concurrent edits shall not silently overwrite newer information when conflict detection is required.	The application detects or safely resolves the defined simultaneous-edit cases and informs users when manual resolution is needed.	Two-session concurrent-edit tests on a task or other shared record.
NFR-REL-06	Must	The application shall handle expected errors without corrupting data or terminating the full application process.	Main error paths return a controlled response, log the technical failure, and preserve database consistency.	Error-path tests and log review.
NFR-REL-07	Should	Application availability should be measured during a stated test or pilot window.	Target ≥99.5% for a defined pilot/monitoring window, excluding planned maintenance; if no meaningful monitoring window exists, report availability as Not Measured.	Health checks or provider monitoring, with time window and failures documented.
NFR-REL-08	Should	A backup and recovery procedure shall be documented for the chosen database/hosting environment.	At least one backup and restore rehearsal is completed in a controlled environment before final handover, where the environment supports it.	Attach the procedure and record the restore verification result.
NFR-REL-09	Should	Important history such as client approvals and change-request decisions shall be retained according to defined rules.	A later deliverable version or new decision does not silently erase prior review history.	Version/review lifecycle tests.
NFR-REL-10	Should	Background reminders and automation shall avoid unintended repeated actions.	Reprocessing the same eligible event does not create duplicate side effects beyond the documented behavior.	Repeat-trigger tests for reminders and rule-based automation.

8. Compatibility and Interoperability Requirements

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-COMP-01	Must	The application shall work in current stable desktop versions of Microsoft Edge, Google Chrome, and Mozilla Firefox, subject to available testing resources.	Critical user journeys pass in the selected supported browsers.	Cross-browser smoke-test checklist with browser versions recorded.
NFR-COMP-02	Must	The responsive web interface shall support common mobile and tablet screen sizes.	Core views and actions remain usable at agreed responsive breakpoints.	Responsive layout tests for navigation, forms, Kanban alternatives, and detail views.
NFR-COMP-03	Must	Date/time behavior shall be consistent and explicit.	Saved timestamps and displayed dates follow the configured time-zone rule; timer calculations do not depend only on the browser clock.	Test date boundaries, time-zone configuration, and timer persistence.
NFR-COMP-04	Should	Data export shall use a documented, commonly usable format for each export type included in scope.	An exported file opens in a compatible tool and matches the filtered records.	Open and reconcile sample exports.
NFR-COMP-05	Should	The solution shall use supported versions of its .NET runtime and main dependencies at implementation time.	Runtime and dependency versions are documented; known blocking compatibility issues are not knowingly left unresolved.	Review project files, package versions, and build output.

9. Maintainability and Testability Requirements

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-MAINT-01	Must	The application shall maintain clear separation between presentation, business logic, and data access.	The solution follows the agreed `TaskFlow.Web`, `TaskFlow.Services`, and `TaskFlow.Data` boundaries, with business rules kept out of views where practical.	Architecture/code review.
NFR-MAINT-02	Must	Dependencies shall be managed through dependency injection where appropriate.	Controllers and services use injected abstractions for core dependencies rather than hard-coded service construction.	Code review and unit-test setup review.
NFR-MAINT-03	Must	Core business rules shall be testable independently of the UI.	Unit tests exist for critical status transitions, timer calculations, permission decisions, progress calculations, and change/review rules where implemented.	Test project and executed test results.
NFR-MAINT-04	Must	The repository shall include reproducible setup and configuration instructions.	A team member can follow the README/setup guide to configure and run the application in the documented environment without undocumented manual steps.	Fresh-clone setup rehearsal.
NFR-MAINT-05	Must	Important errors shall be logged using a consistent approach.	Logs contain sufficient diagnostic context for failures without writing passwords, tokens, or unnecessary sensitive data.	Trigger representative failures and inspect sanitized logs.
NFR-MAINT-06	Should	Public APIs or important service contracts shall be documented.	Relevant request/response contracts, events, and setup assumptions are documented where needed for cross-module integration.	Documentation review against implemented interfaces.
NFR-MAINT-07	Must	Code changes shall use the agreed version-control and review workflow.	Significant work is committed in focused branches/PRs or another agreed reviewable Git workflow; secrets are excluded.	Review repository history and merge records.

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-MAINT-08	Should	The team shall keep implementation and documentation synchronized.	Features deferred, changed, or completed are reflected in the backlog, requirements, and user/setup documentation.	Compare GitHub Issues, requirements, and release notes.

10. Accessibility Requirements

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-ACC-01	Must	Core controls shall be operable by keyboard where applicable.	Users can reach and activate core navigation, forms, dialogs, and task actions with keyboard input; focus is visible.	Keyboard-only walkthrough of critical journeys.
NFR-ACC-02	Must	Form controls shall have understandable labels and error messages.	Each core form input has a programmatic or visible label and validation feedback is associated with the relevant field where feasible.	Accessibility inspection and form tests.
NFR-ACC-03	Should	Text and interactive states should meet WCAG 2.1 AA contrast criteria where applicable.	Body text and important controls meet the relevant contrast target where measured; exceptions are documented.	Contrast checker and visual inspection.
NFR-ACC-04	Should	Charts, progress indicators, and status colors shall have text or alternative summaries.	A user is not required to distinguish color alone to understand status or progress.	Inspect charts/badges and test with grayscale or assistive descriptions where feasible.
NFR-ACC-05	Should	Dynamic feedback should be understandable to assistive technology where practicable.	Key success/error updates and live status changes have meaningful labels or announcements where supported by the UI implementation.	Accessibility review of dynamic interactions.

11. Privacy and Data Handling Requirements

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-PRIV-01	Must	Users shall only see personal and project data necessary for their role and authorized scope.	Access-control tests show no unauthorized exposure of another user's private tasks, workspace data, or client project content.	Cross-account and cross-project access tests.
NFR-PRIV-02	Must	Internal team information shall be separated from client-visible information.	Internal-only comments, files, time records, and notes are not shown to clients unless explicitly approved for sharing.	Client-role tests for every shared content type.
NFR-PRIV-03	Must	Test/demo data shall not contain real private client information without authorization.	The demonstration uses synthetic or approved sample data; secrets and private customer files are not committed to source control.	Repository and demo-data review.
NFR-PRIV-04	Should	Data retention, export, and deletion behavior shall be documented for records covered by the implementation.	The team documents which records are archived, retained, exported, or deleted and any history retained for auditability.	Review user guide and data-handling decisions.

12. Operational and Deployment Requirements

ID	Priority	Requirement	Proposed measurable target	Verification method
NFR-OPS-01	Must	The application shall support a documented local setup and execution path.	A clean setup can build, apply migrations, and start using the documented configuration.	Fresh-clone setup test.
NFR-OPS-02	Should	Environment-specific configuration shall be separated from source code.	Database credentials, email settings, and deployment secrets are provided through environment-specific configuration rather than hard-coded values.	Inspect source and deployment settings.
NFR-OPS-03	Should	Deployment should expose a basic health/diagnostic check where hosting allows it.	The team can verify whether the application is running and identify a failed dependency without exposing sensitive detail.	Health-check test or documented equivalent.
NFR-OPS-04	Should	The team shall keep a deployment/recovery checklist.	The checklist describes configuration, migrations, startup verification, and recovery steps for the chosen environment.	Review checklist and perform a rehearsal where feasible.
NFR-OPS-05	Could	CI/CD may automate build and test checks on GitHub changes.	The pipeline, if included, reports build/test success or failure and does not deploy secrets from untrusted branches.	Execute pipeline with a success and failure example.

13. Measurement and Reporting Rules

1. Each measured target must specify the environment, build, dataset, sample size, test date, and method.
2. Do not report a proposed target as an achieved result. Use `Pass`, `Fail`, `Partially met`, or `Not measured` with evidence.
3. Performance tests should use representative data and repeat requests; report percentile results where possible instead of a single best-case measurement.
4. Security acceptance requires negative permission testing, not only successful user flows.
5. If hosting or monitoring is unavailable, mark cloud availability or deployment-dependent items as `Not measured` and explain the limitation.
6. Any requirement deferred from the agreed release must be recorded in the backlog and reflected in the final report.

14. Suggested Requirement Owners

Area	Primary owner	Supporting members
Authentication, personal dashboard, navigation, notifications foundation, repository and deployment coordination	Mostafa Mohamed Mahmoud Mohamed	All members
Workspace/project permissions and membership isolation	Ahmed Reda Ahmed Sedik	Mostafa, Eman
Task operations, Kanban, comments, attachments, task history and handover	Alaa Essam Hamed Abdelkader	Ahmed, Ahd, Eman
Performance measurement support, timer correctness, Gantt/calendar, reports, progress and health-alert rules	Ahd Ammar Eisa El-Baqary	All module owners where data is shared
Client data isolation, deliverable review, client-facing content and change-request privacy	Eman Mohamed Masad El-Metwally	Ahmed, Alaa, Ahd, Mostafa
Cross-cutting security, usability, reliability, and accessibility	All members	Mostafa coordinates review and evidence collection

Public marketing pages are handled separately by Mostafa and are excluded from the internal application-module allocation above.

15. Sign-off

Reviewer	Name	Status	Date / Comments
Team Lead	Mostafa Mohamed Mahmoud Mohamed	Pending review	
Workspace / Project Module	Ahmed Reda Ahmed Sedik	Pending review	
Task / Kanban Module	Alaa Essam Hamed Abdelkader	Pending review	
Scheduling / Reporting Module	Ahd Ammar Eisa El-Baqary	Pending review	
Client / Deliverables Module	Eman Mohamed Masad El-Metwally	Pending review	
DEPI Lecturer / Supervisor	To be completed	Pending feedback	

Document note: This is a proposed NFR specification based on the TaskFlow project brief, common web-application quality considerations, and the team's agreed scope. The team and lecturer should review the proposed targets before treating them as final acceptance thresholds. No implementation or test result is implied by the presence of a requirement.